A Timeline of AI and Machine Learning

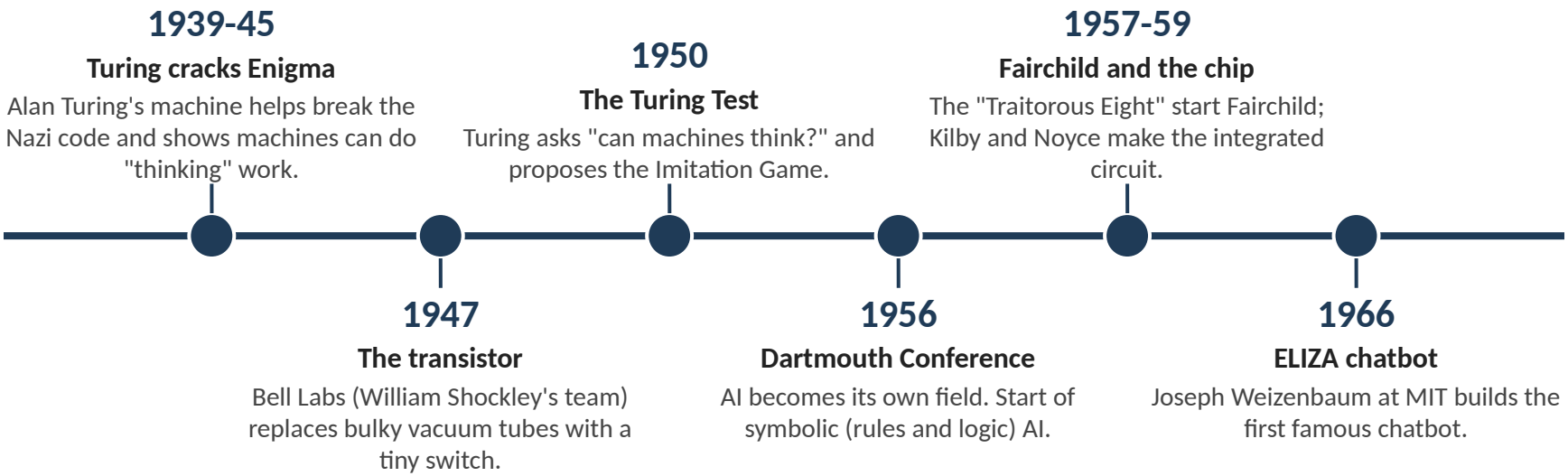
From Turing's codebreaking to ChatGPT (1939 to 2024)



Pavan Kumar Seethamsetty
AIML-500-01N1Z Machine Learning Fundamentals
Indiana Wesleyan University

1939 to 1971: The Foundations

Codebreaking, chips and the first chatbot



The Two AI Winters

When the hype got ahead of what computers could actually do

First AI Winter (about 1974 to 1980)

Causes: Big promises of thinking machines didn't come true. The 1973 Lighthill Report (UK) said AI hit a "combinatorial explosion" and computers were too slow with too little memory.

Impact: DARPA cut general AI funding in 1974. Labs shrank and researchers moved to other fields.

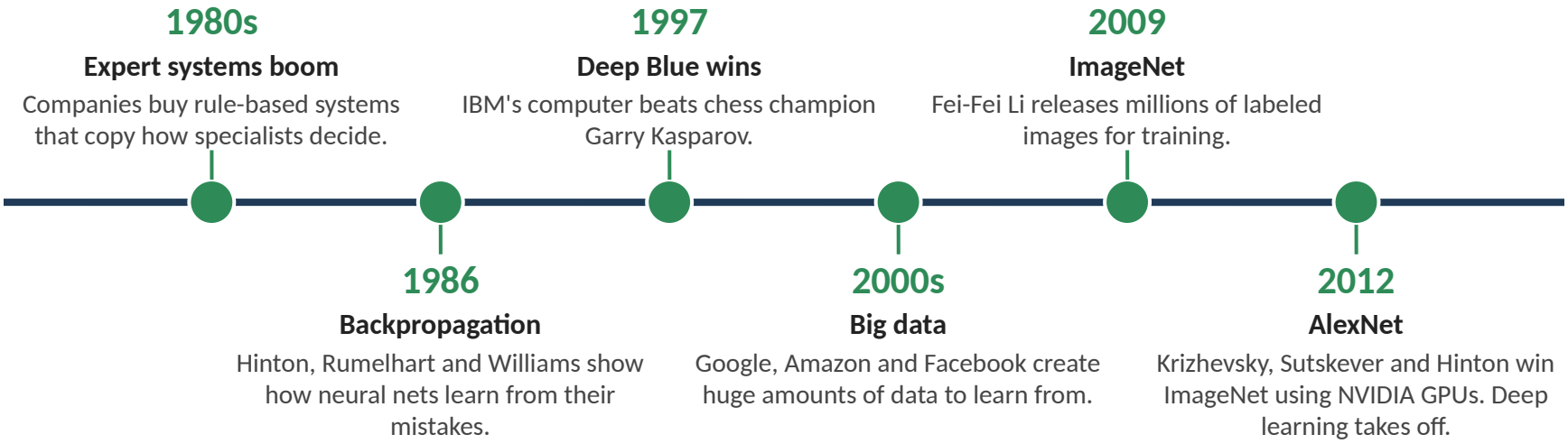
Second AI Winter (about 1987 to early 1990s)

Causes: Expert systems were expensive and hard to update. Cheaper PCs killed the Lisp machine market, and Japan's Fifth Generation project fell short.

Impact: Another round of funding cuts, but a few people like Geoffrey Hinton kept working on neural networks.

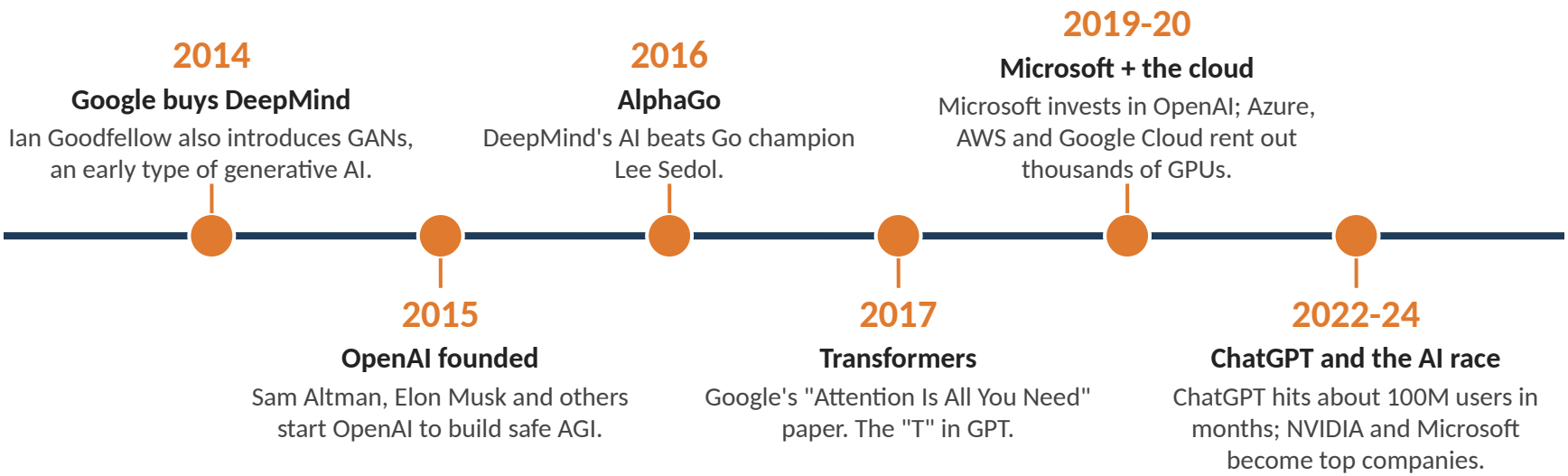
1980 to 2012: Learning From Data

Neural networks come back once data and hardware catch up



2014 to 2024: Cloud and Generative AI

Big tech and new labs race to build bigger models



What I Learned

Three ideas that stood out while making this timeline

1

Narrow AI vs. AGI

Everything here, from Deep Blue to ChatGPT, is narrow AI (good at specific tasks). AGI, which could reason about anything like a person, doesn't exist yet.

2

Hardware matters

Backpropagation was around in 1986, but it needed big data, GPUs and the cloud to work. Training compute has doubled about every 6 months since 2010.

3

Predictive to generative

Older machine learning predicted things (tag a photo, flag fraud). Generative AI creates new text, images and audio.

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